

Movie Success Analysis: A Machine Learning Study on Factors Affecting Movie Revenue.

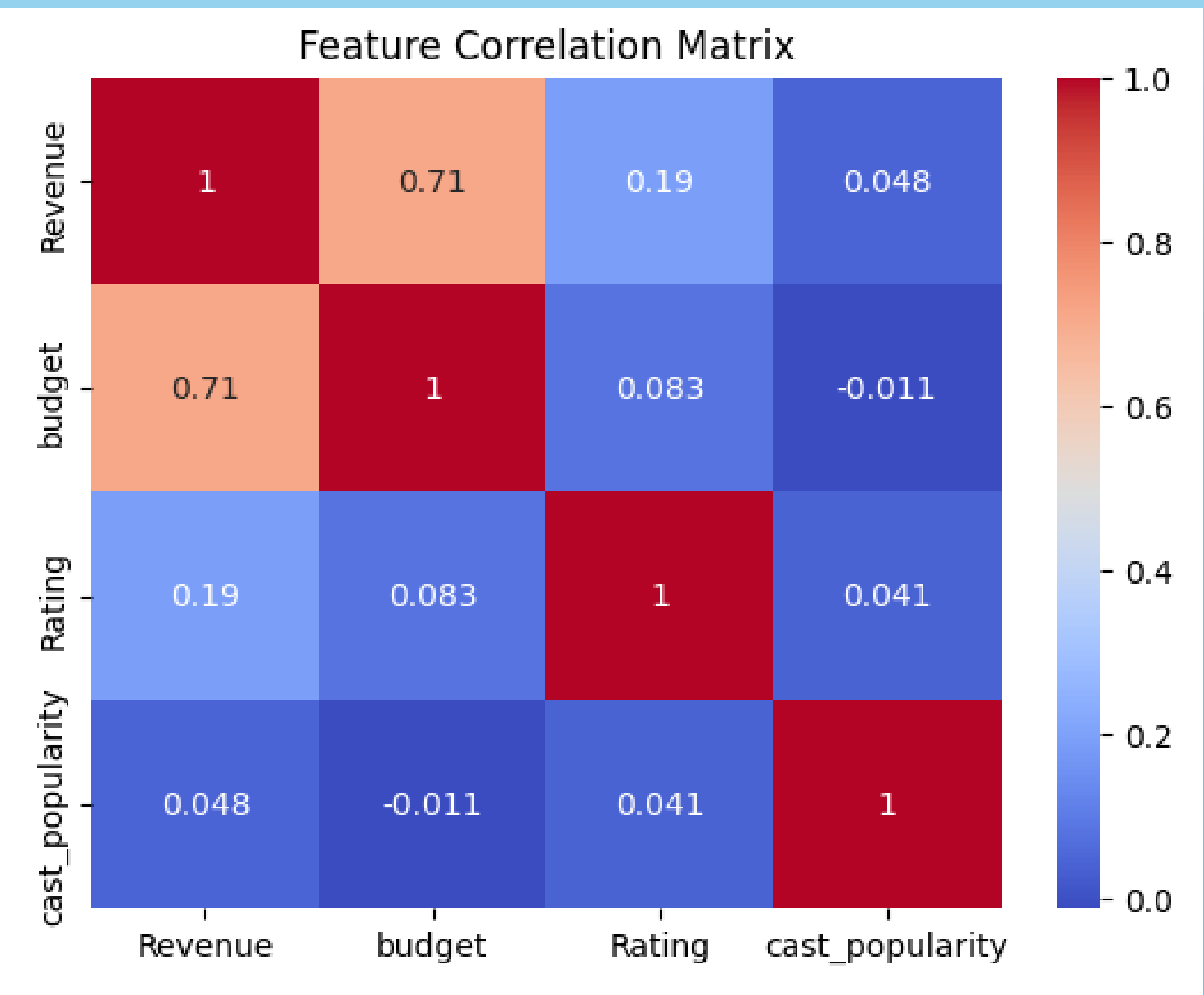
PROBLEM STATEMENT

It is unclear how significant the impact of each revenue-affecting factor is on box office success. Additionally, some factors include distinct categories with different patterns that affect movie revenue, such as audience ratings and critics' ratings.

Objectives

This project will compare the effect of each factor on box office success across genres by analyzing the correlation between each factor and movie revenue and detect patterns across factors.

Dataset Overview: Dataset name: Movies Success Factors Dataset. • Sources: Box Office Mojo, IMDb, and TMDb. • Number of records: 5400 records. • Number of features: 13 features. • Description: Each row contains the movie title, year it was released in, audience rating obtained from IMDb, critics’ rating, season it was released in, budget, genres, an engineered cast popularity score, an indicator for whether the movie had a critics’ rating or not, log-scaled revenue and budget, and a revenue class of high or low. • Key variables: Revenue, ratings, season, budget, cast popularity, and revenue class.



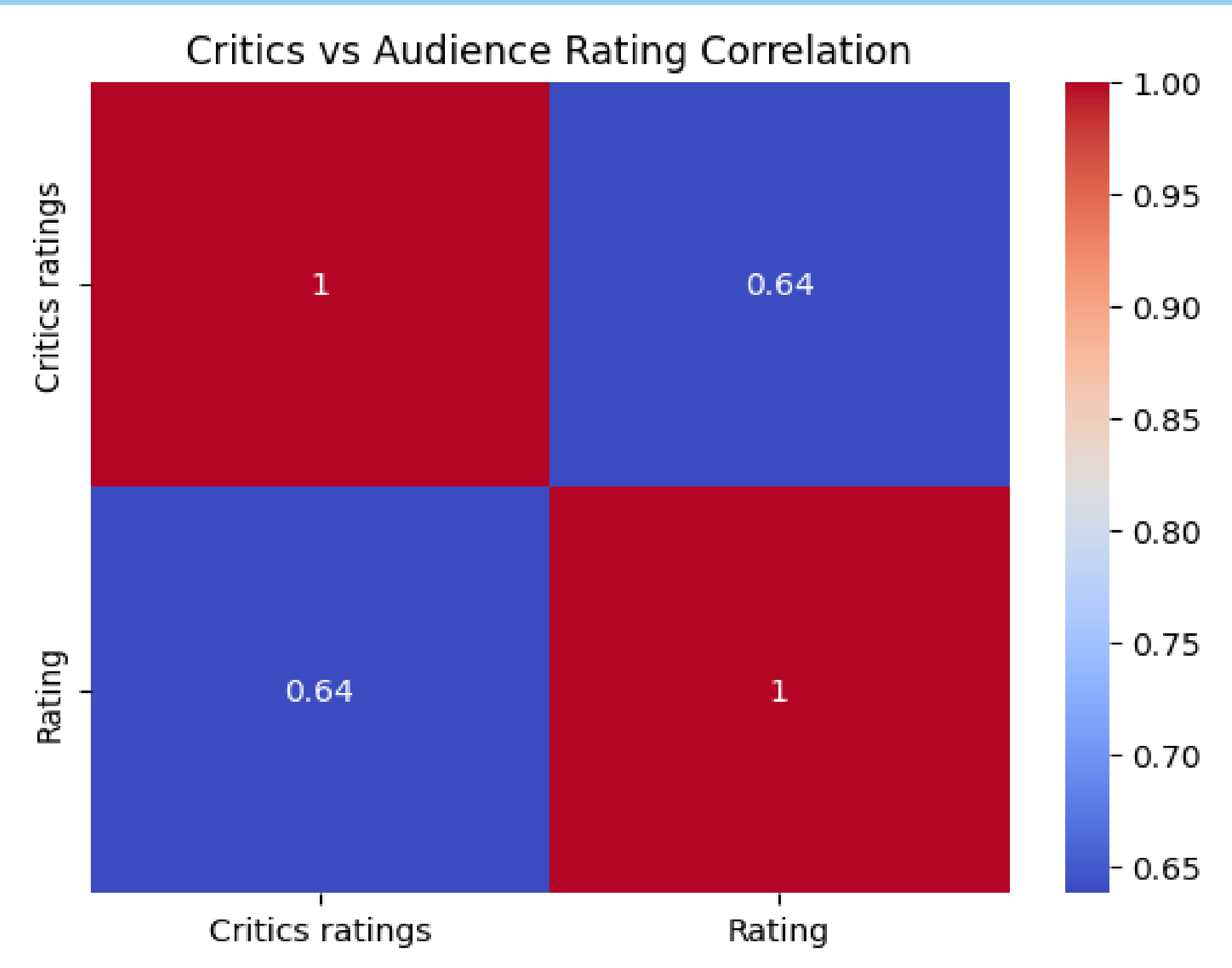
A heatmap representing the correlation between the movies’ budget, rating, cast popularity, and revenue was used to identify which factors directly affect revenue the most. Through the analysis, it was concluded that budget affects revenue the most due to the high coefficient of correlation between them.

Key Findings

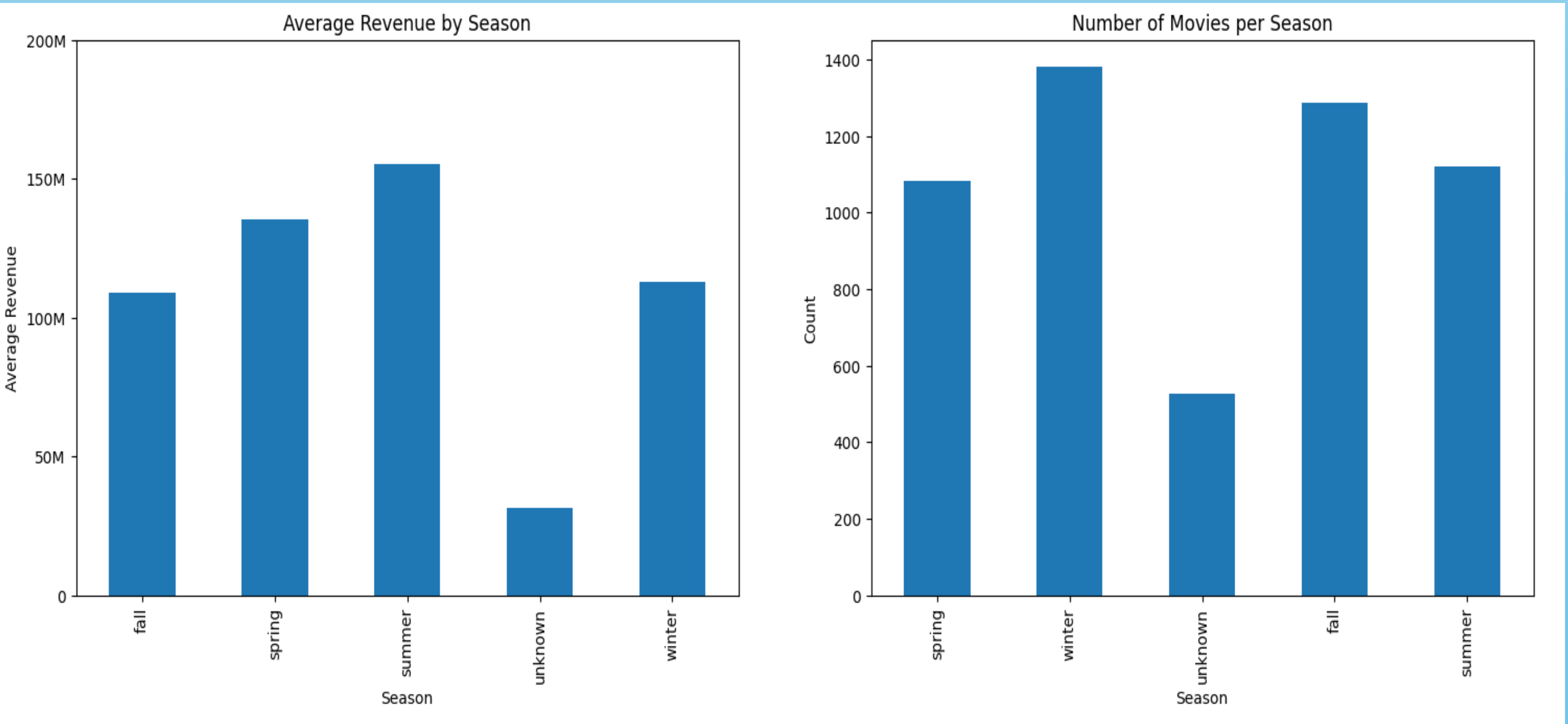
- Budget was the strongest indicator of revenue, with a coefficient of correlation of 0.71.
- Ratings have a weak effect on movie revenue. Although some highly rated movies have high revenues, IMDb ratings and movie revenue had a coefficient of correlation of 0.19.
- Seasons have a mild but noticeable effect on movie revenue; summer movies have higher average revenue than other seasons. Additionally, summer movies have higher median revenue and more consistent performances. In spite of having the greatest number of high-revenue movies, winter had more variability and failures than any other season.
- Critics and audiences tend to agree on the rating of most movies, with a coefficient of correlation of 0.64.

Recommendations

- Movie producers should prioritize movie budget over other revenue-affecting factors.
- Marketing strategies should be focused on visibility rather than aiming for higher ratings since ratings have a weak effect on movie revenue.



A heatmap representing the correlation between the critics’ ratings and audience ratings was used to visualize the strength and direction of the relationship between them. Since the correlation coefficient was 0.64, there was a strong agreement between the critics and the audiences for most of the movie’s ratings.



Average revenue by season: A bar graph of the average revenue by each season was used to avoid the unequal group bias of a total revenue bar graph and visualize which season had better-performing movies. The graph showed that summer movies had the highest average revenue, indicating that movies released in the summer usually perform better than those released in any other season. Number of movies per season: A bar graph was used to find the number of movies for each season collected in the dataset. This acted as a preliminary visualization before making conclusions on the revenue by season since some seasons may have more movies. All of the seasons had around the same number of movies, which ensured data balance.